nelsonnet

WORKSHEET

Chapter 3: Optimisation – Area and volume

What animal has four noses?

4	6	2		7	3	5	1

Questions

1 A 60 m length of fencing is to be made into the boundary of a rectangular paddock. What is the maximum area it can enclose?

2 A square-based closed box has a total surface area of 5400 cm^2. What is the maximum volume it can contain?

3 A ribbon is to be tied lengthwise and then crosswise around a square-based rectangular present leaving an extra 30 cm length for a bow. If the volume of the present is 2592 cm^3, what is the minimum length of ribbon required?

4 A new paddock is to be formed alongside an already fenced one. 80 m of fencing is to be used to form the length and base of the right-angled triangular paddock. What is the maximum area that can be enclosed?

5 Find the minimum surface area of a cylinder with a volume of 300 cm^3.

6 Find the maximum volume of a cylinder whose height and radius do not exceed a combined total of 12 cm.

7 A square-based rectangular prism is required to hold 1000 m^3. What is the minimum surface area of the prism?

Solutions

E 27000 cm^3

G 225 m^2

H 804.3 cm^3

L 138 cm

S 600 m^2

T 800 m^2

U 248.1 cm^2

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